NC STATE UNIVERSITY

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CSC 491/ CSC 525/ ECE 525

NNDL - 20

Deep Learing based Multi-class Network Intrusion Detection

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Problem Statement

Designing and developing a machine learning-based Network Intrusion Detection System capable of:

- **Classifying 20+ distinct network anomalies** in real-world traffic
- Effectively handling **highly imbalanced datasets**
- Capturing the **sequential patterns** inherent in network traffic

Data

Dataset Used

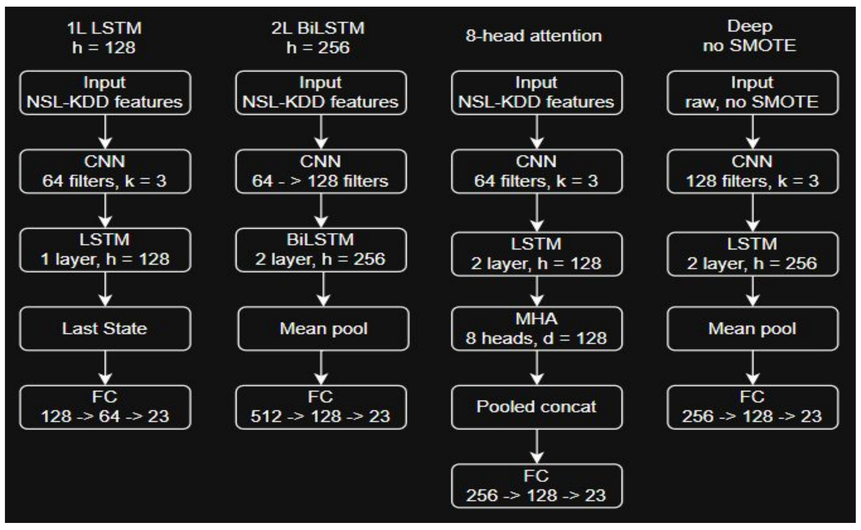
- NSL-KDD dataset from Kaggle, benchmark for network intrusion detection research

Input Features

- 41 tabular features per connection
- Connection details (e.g., source/destination IP, port)
- Payload content attributes
- Temporal traffic patterns

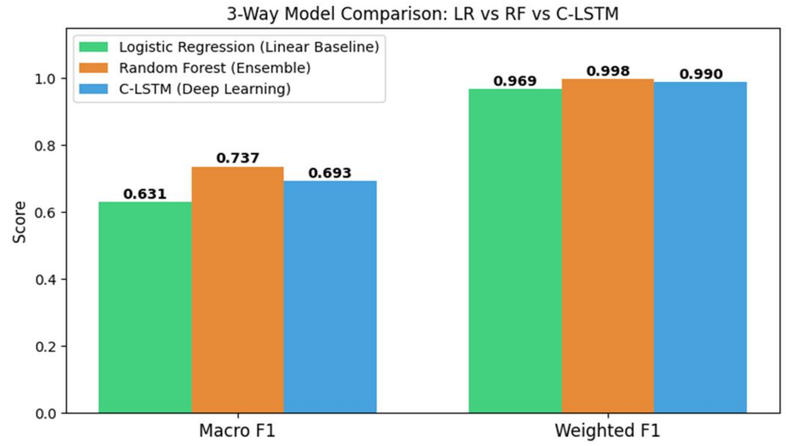
Approach and Network Architecture

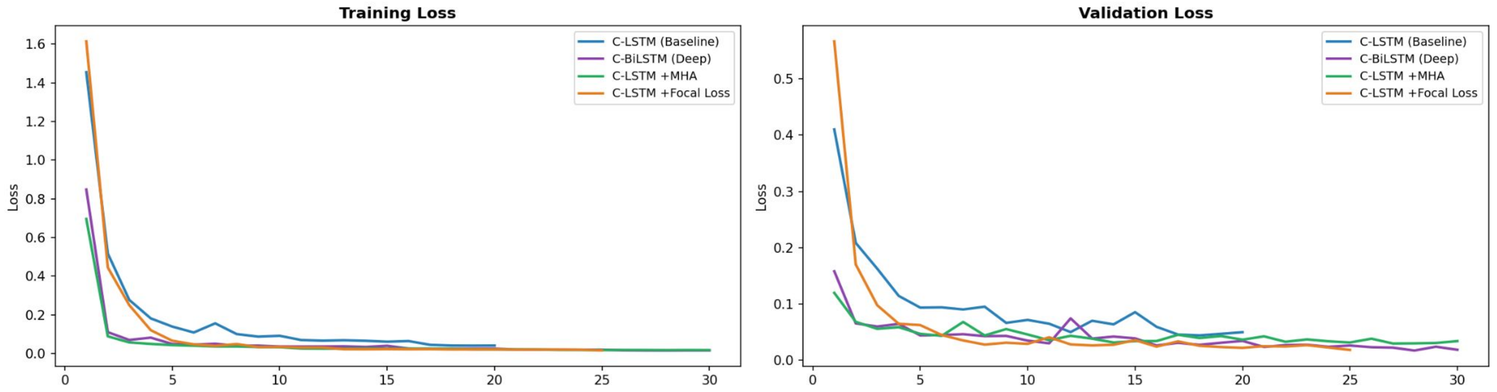
- 4 models with shared CNN → LSTM → FC layer backbone (Chosen based on baseline evaluations over LR and RF).
- CNN for local feature extraction, LSTM for capturing sequential dependencies and FC layer to map to 23 attack classes.
- Each iteration introduced specific changes from the C-LSTM baseline. Starting from bidirectionality and mean pooling to self attention and finally focal loss to target class imbalance better.

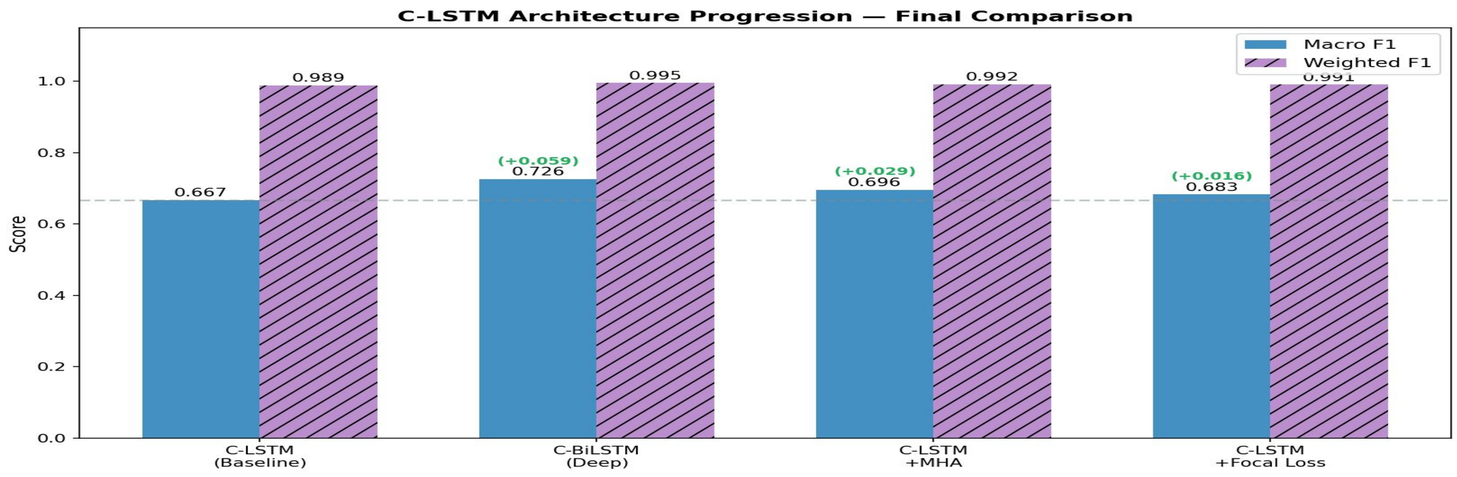


Evaluation Plan and Results

- **Split:** 90/10 train/val · KDDTest+ as held-out test · SMOTE on train only (no leakage)
- **No Accuracy:** 13,469 normal vs. 1 spy sample — majority-class bias makes it meaningless
- **Macro F1 (Primary):** Equal weight across all 23 classes — penalises ignoring rare attacks
- **Weighted F1 (Secondary):** Context only — dominated by normal, masks minority failure
- **Per-Class:** Precision → minimise false alarms · Recall → catch rare stealthy attacks
- **Visuals:** Confusion matrices (row-normalised) · Loss curves · 3-model comparison chart







Main References

[1] Kaggle NSL-KDD dataset for Network Security, Information Security and Cyber Security
https://www.kaggle.com/datasets/hassan06/nslkdd/data

[2] KDD Cup 1999 Dataset, https://archive.ics.uci.edu/ml/datasets/kdd+cup+1999+data